

# homework2

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## 1 Homework 2

study number: s234857

```
[17]: import numpy as np
```

### 1.1 Multiple choice

A) 3,

B) 3,

C) 2,

D) 2

#### 1.1.1 3.1)

We have that  $p_0(x) = c_0 = 1$ . Then,

$$p_1(x_1) = 1.105 = 1 + c_1(0.05 - 0) \quad (1)$$

(2)

$$\implies c_1 = 0.105 \cdot 20 = 2.1, \quad (3)$$

and

$$p_2(x_2) = 1.221 = 1 + 2.1(0.1 - 0) + c_2(0.1 - 0)(0.1 - 0.05) \quad (4)$$

(5)

$$= 1.21 + 0.005c_2 \quad (6)$$

(7)

$$\implies c_2 = 2.2 \quad (8)$$

So the interpolating polynomial  $p_2$  on Newton form of is expressed as

$$p_2 = 1 + 2.1(x) + 2.2(x)(x - 0.05) \quad (9)$$

$$= 1 + 2.1((x) + 2.2(x - 0.05)) \quad (10)$$

## 1.2 3.2)

From the chain rule we know that  $\frac{d^n}{dx^n}e^{ax} = a^n e^{ax}$  and thus  $\frac{d^{n+1}}{dx^{n+1}}e^{ax} = a^{n+1}e^{ax}$ . We therefore define  $M := \max_{x \in [0, 0.1]} 2^{n+1} f(x) = 2^{n+1} \cdot 1.221$

Furthermore, the spacing between the nodes is found as  $h = \frac{0.1-0}{n} = \frac{0.1}{n}$

Inserting into the second interpolation theorem, we have that

$$|e^{2x} - p_n(x)| \leq \frac{1.221 \cdot 2^{n+1}}{4(n+1)(10n)^{n+1}} \quad (11)$$

$$(12)$$

$$= \frac{1.221 \cdot 2^{n+1}}{4(n+1)(5n)^{n+1}2^{n+1}} \quad (13)$$

$$(14)$$

$$= \frac{0.3053}{(5n)^{n+1}(n+1)} \quad (15)$$

Which is now found numerically for :

```
[11]: M = lambda n: 2**(n+1) * 1.221
h = lambda n: 0.1 / n
bound = 1e-5
for n in range(1, 100):
    error = 0.3053 / ((5*n) ** (n+1) * (n+1))
    if error < bound:
        print(n, error)
        break
```

3 1.5076543209876545e-06

Hence  $n = 3$  is the smallest  $n$  such that the error bound is less than  $10^{-5}$  is

## 1.3 The composite Simpson's rule

1) The MySimpson function

```
[14]: def MySimpson(f, a, b, n):

    if n % 2:
        print("n must be even")
        return float("nan")
    h = (b - a) / n
    f_evaluations = [f(a + i * h) for i in range(n + 1)]

    for i in range(1, n, 2):
        f_evaluations[i] *= 4
    for i in range(2, n-1, 2):
        f_evaluations[i] *= 2

    return h / 3 * sum(f_evaluations)
```

## 2) Examining error of the Simpson's rule

Firstly the true integral is evaluated:

$$\int_0^{\pi} \sin(x) dx = -\cos(\pi) + \cos(0) = 2$$

We now wish to observe how whether the true error term is proportional to the expected  $n^{-4}$ . It is expected that  $\frac{|E_n|}{n^4}$  should converge to some constant.

Testing for even  $n$  from 2 to 50:

```
[35]: def maxSimpsonError(f4_bound,a,b,n):
        return (b - a)**5 / (180 * n**4) * f4_bound

F_true = 2
absolute_error = []
relative_error = []
n_max = 50
for n in range(2, n_max, 2):
    absolute_error.append(abs(F_true - MySimpson(np.sin, 0, np.pi, n)))

n_values = np.arange(2, n_max, 2)
convergence_test = np.array([
    absolute_error[i] * n_values[i]**4 for i in range(len(absolute_error))
])
print("convergence test")
print(convergence_test)
```

convergence test

```
[1.51032164 1.16729728 1.11869382 1.10252011 1.09517315 1.09121834
 1.08884587 1.08731092 1.08626076 1.08551069 1.08495631 1.08453499
 1.0842073  1.08394742 1.08373784 1.08356637 1.08342429 1.08330525
 1.08320453 1.08311855 1.08304456 1.08298044 1.08292451 1.08287543]
```

It is observed that the error is proportional to approximately  $1.0829n^{-4}$

- 4) From the chain rule we know that  $\frac{d^n}{dx^n} e^{ax} = a^n e^{ax}$ , thus  $f^{(4)} = 16e^{2x}$ . As  $e^{2x}$  is monotonically increasing, we find its maximum value on the interval at  $f^{(4)}(0.1) = 16 \cdot 1.221 = 19.536$ .

As the bound on the maximum error is of interest, we get that

$$|E_t| \leq \max_{\xi \in I} f^{(4)}(\xi) \frac{(0.1 - 0)^5}{180n^4} \quad (16)$$

$$= \frac{19.536 \cdot 10^{-5}}{180n^4} \quad (17)$$

$$\Rightarrow n^4 \geq \frac{19.536 \cdot 10^{-5}}{180|E_t|}. \quad (18)$$

Hence for  $E_t = 10^{-5}$ ,

$$n^4 \leq \frac{19.536}{180} \tag{19}$$

$$\Rightarrow n \leq \left( \frac{19.536}{180} \right)^{\frac{1}{4}} \tag{20}$$